

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

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Department of Computer Science and Engineering

SOFTWARE REQUIREMENTS SPECIFICATION SQUIGGLE

A Real-Time Multiplayer Drawing and Guessing Game

SWE 4404 — Software Project Lab–II | Academic Year 2024–25

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Repository	https://github.com/zahin002/Squiggle
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The headings use Word/Google Docs heading styles for automatic contents generation after import. The acceptance criteria in Section 5 are compact restatements of the visible screenshot cells; they are not a verbatim spreadsheet export.

1. Introduction

1.1 Purpose

This Software Requirements Specification describes what Squiggle is expected to do, who will use it, and how its requirements can be checked. It brings together the team's SRS presentation, Google Form stakeholder responses, requirements spreadsheet screenshots, project proposal, and acceptance-criteria review documents.

1.2 Product scope

Squiggle is a browser-based social game where a designated player draws a secret word and others guess within a time limit. Players can create or join rooms, choose settings, and play without registering as guests. The specified scope also includes word modes, shared canvas events, player progression, spectators, voice chat, and optional power-ups.

1.3 Document conventions

FR_1–FR_24 identify functional requirements. US_1–US_30 identify user stories. High/Medium/Low denote priorities; "Done" and "Not Done" refer to the team's later completion reporting, not acceptance-test certification. Requirements phrased with "shall" are intended system behaviors; verification examples are test conditions, not automatically verified results.

2. Overall Description and Stakeholders

2.1 User roles

Role	Primary need
Guest player	Join and participate without a permanent account
Registered player	Sign in and retain rewards, rank, and avatar
Host	Create/manage room and settings
Drawer	Select a word and sketch it on a shared canvas
Guesser	See the drawing, enter guesses, and earn points
Spectator	Observe live gameplay without changing the result

2.2 Product environment

The proposed system uses a React web interface, Express-based REST APIs, Socket.IO for multiplayer messages, MongoDB/Mongoose for persistent information, and WebRTC for optional browser-based voice. An in-memory room store handles rapidly changing game state. This section describes the architecture reflected in the provided project records; the SRS focuses on expected behavior rather than claiming that each feature has passed testing.

3. Requirements Elicitation: Google Form

3.1 Survey method and response base

The provided Google Forms response export contains 19 dated submissions from 28–29 June 2026. The form invited social, casual, competitive, and drawing-game-oriented participants to describe preferred onboarding, drawing tools, room settings, rewards, game modes, spectator use and voice communication. These are exploratory responses, not a representative market sample.

3.2 Questions collected

1. Which best describes you as a player?

2. When trying a new game for the first time, what do you prefer?

3. As a room host, which settings matter most to you?

4. Which drawing tools are essential for you?

5. If you did not know what a word looked like, would you use a private drawer reference-image search?

6. Which word categories would you enjoy most?

7. Would you enjoy the Wordle-inspired blind-guessing/drawing mode?

8. When guessing, which features would improve your experience?

9. Should earlier correct guesses earn more points?

10. Which rewards would motivate you to keep playing?

11. How do you feel about spending earned coins on power-ups?

12. Should a voted-out player lose coins as a penalty?

13. How important is real-time voice chat during a game?

14. Would you want to join as a spectator without playing?

15. What single feature would make Squiggle your favorite drawing game?

3.3 How responses informed requirements

Observed response themes	Requirement implications
Some respondents prefer guest access; others request one-click Google login	FR_1–FR_4: registration, authentication, daily login and guest access
Hosts request player-count, time, genre, private rooms, and custom words	FR_5–FR_6: configurable room setup and word selection
Brush, eraser, undo/redo, shapes and color picker mentioned repeatedly	FR_7–FR_8: drawing controls and synchronization
Players request clues, word-length hints and speed-related scoring	FR_9, FR_17, FR_23–FR_24
Interest in reward systems, power-ups and avatars varies	FR_10–FR_13, FR_18–FR_21
Respondents consider spectator participation and microphone controls	FR_14–FR_15

A suggested reference search also appears in the survey, but FR_16 was later reported as not completed. The survey records user interest; it does not establish successful feature delivery.

4. Functional Requirements (FR_1–FR_24)

The identifiers and story mapping follow the original Google Sheet screenshots. Requirement statements below are concise, report-ready paraphrases. Priorities are broad groupings where the screenshot text is not fully legible; consult the source sheet for exact row-level priority values.

ID	User story	Functional requirement	Priority*	Final team status
FR_1	US_1–US_2	Register a unique account and prevent duplicate usernames	High	Done (reported)
FR_2	US_3	Authenticate registered users with username/password and JWT	High	Done (reported)
FR_3	US_4	Provide first daily-login coin reward	Low	Done (reported)
FR_4	US_5	Allow guest access without registration	High	Done (reported)
FR_5	US_6–US_7	Create configurable rooms, including private rooms	High	Done (reported)
FR_6	US_8–US_9	Provide generated word	Medium	Done (reported)

		choices and genre selection		
FR_7	US_10	Support a drawing canvas with drawing tools	High	Done (reported)
FR_8	US_11	Synchronize drawing events in real time	High	Done (reported)
FR_9	US_12	Provide live chat, guessing and result feedback	High	Done (reported)
FR_10	US_13	Calculate and distribute game rewards	High	Done (reported)
FR_11	US_14	Show a global top-player leaderboard	High	Done (reported)
FR_12	US_15	Show the user's reward history	Medium	Done (reported)
FR_13	US_16	Allow valid in-game power-up purchases	Medium	Done (reported)
FR_14	US_17–US_18	Support spectator entry and spectator chat	Medium	Done (reported)
FR_15	US_19	Provide optional WebRTC voice communication	High	Done (reported)
FR_16	US_20	Provide private drawer-only reference image search	Medium	Not Done
FR_17	US_21–US_23	Provide the Wordle-inspired game mode and phases	High	Done (reported)
FR_18	US_24	Provide Hint Steal power-up	Medium	Done (reported)
FR_19	US_25	Provide Extra Clues power-up	Medium	Not Done
FR_20	US_26	Provide Freeze Time power-up	Medium	Done (reported)
FR_21	US_27	Allow avatar customization	Medium	Done (reported)
FR_22	US_28	Provide vote-kick moderation	Medium	Done (reported)
FR_23	US_29	Show live in-game standings	High	Done (reported)
FR_24	US_30	Award time-sensitive guess scores	High	Done (reported)

*Priority labels are summary interpretations from the proposal and priority-audit discussion, not a verbatim reproduction of all 30 spreadsheet cells.

5. User Stories and Acceptance Criteria (US_1–US_30)

The criteria below use the detailed requirement screenshots and audit notes. In a few cases the screenshot truncates a long cell; such criteria are summarized for readability. To reproduce every bullet exactly, attach/export the full editable Google Sheet.

5.1 User stories US_1–US_10

Story / FR	User need	Acceptance criteria (condensed)
US_1 / FR_1	Register account	Unique username of 3–20 characters; password stored as a hash; a successful registration returns an authenticated session.
US_2 / FR_1	Prevent duplicate accounts	Existing username is rejected and registration does not create duplicate user.
US_3 / FR_2	Log in	Valid credentials return JWT and profile; invalid credentials show an error.
US_4 / FR_3	Daily login reward	Award 2 Gold Coins once per day per registered user; no duplicate claim on same date.
US_5 / FR_4	Play as guest	Guest can join without registering; guest

		progression is not permanent.
US_6 / FR_5	Create a room	Host chooses player count, time, rounds and mode; generated room code is six characters.
US_7 / FR_5	Private room	Private room requires the code to enter; room settings remain retrievable by ID.
US_8 / FR_6	AI word selection	Offer three word choices; use an appropriate fallback if AI is unavailable.
US_9 / FR_6	Genre selection	Host chooses genres; generated/selected words respect enabled categories.
US_10 / FR_7	Canvas tools	Brush/eraser, undo/redo, clear, colors and geometric shapes can be used while drawing.

5.2 User stories US_11–US_20

Story / FR	User need	Acceptance criteria (condensed)
US_11 / FR_8	Live canvas updates	Other room participants receive strokes in correct order with no deliberate page refresh.
US_12 / FR_9	Live chat and guesses	Players submit guesses; correct entries receive distinct feedback and scoring.
US_13 / FR_10	Calculate rewards	Correct guesses and results award stated currencies; reward rules are applied consistently.
US_14 / FR_11	Global leaderboard	Display a ranked top-20 list using stored statistics; show the intended player fields.
US_15 / FR_12	Reward history	Registered user can view recent reward transactions with type, amount and date.
US_16 / FR_13	Purchase power-up	Check sufficient coins/diamonds; deduct appropriately; reject insufficient balance.
US_17 / FR_14	Spectator mode	Spectator views game without guessing/drawing; full-room handling obeys spectator rules.
US_18 / FR_14	Spectator chat	Separate spectator-only chat and access control, as originally specified.
US_19 / FR_15	Voice chat	Voice opt-in, microphone mute and room host voice controls are provided.
US_20 / FR_16	Drawer reference image search	Only current drawer may search; results are hidden from guessers. Not implemented in final team report.

5.3 User stories US_21–US_30

Story / FR	User need	Acceptance criteria (condensed)
US_21 / FR_17	Wordle blind phase	Six-attempt grid and color-coded feedback for Wordle-inspired guesses.
US_22 / FR_17	Wordle drawing phase	Drawing unlocks at the specified stage and guesses proceed until time or completion.
US_23 / FR_17	Wordle chat restrictions	A completed correct word cannot be used to compromise other players' remaining guesses.
US_24 / FR_18	Hint Steal	Reveal permitted letters at the specified escalating coin/diamond cost.
US_25 / FR_19	Extra Clues	Provide additional clue selection at defined price. Not implemented in final team report.
US_26 / FR_20	Freeze Time	Apply time extension/freeze under the defined power-up rules and charge the user.
US_27 / FR_21	Avatar customization	Edit avatar appearance and preview it; registered settings can be saved.
US_28 / FR_22	Vote-kick	Allow eligible votes; apply threshold and removal/penalty rules.
US_29 / FR_23	In-game leaderboard	Scores update after turns/rounds and remain visible to active participants.
US_30 / FR_24	Speed-based scoring	Earlier correct guesses earn higher scores based on remaining time.

6. Non-Functional Requirements

Quality attribute	Target / expectation	Verification approach
Performance	Drawing updates target <100 ms under normal network conditions (proposal target, not measured here).	Two-client timestamp measurements
Usability	New player should be able to join a game within 60 seconds (proposal target).	Timed user walkthrough
Security	Passwords hashed; JWT checked on protected endpoints; secret word withheld from guessers.	Authentication and payload inspection
Reliability	Temporary disconnection should allow reasonable recovery.	Disconnect/rejoin simulation
Scalability	Multiple independent rooms without gameplay cross-talk.	Concurrent-room integration testing
Compatibility	Modern browser and responsive interface.	Browser/viewport manual checks
Privacy	Voice requires browser permission and remains optional.	Microphone permission and mute tests

7. Use Cases and Interaction Flows

UC-01: Register / sign in

Actor: Registered player

Basic flow: Enter credentials → submit auth request → validate account → issue JWT → open lobby.

UC-02: Create and join a room

Actor: Host or guest

Basic flow: Select settings → validate → create room/code → player joins using code → room state is published.

UC-03: Play a drawing turn

Actor: Drawer and guesser

Basic flow: Server selects drawer and word → drawer sends drawing events → guessers send answers → score/time update → next turn.

UC-04: Spectate

Actor: Spectator

Basic flow: Select spectator role → join room → receive drawing and optional voice audio; no scoring actions.

UC-05: Use power-up

Actor: Registered player

Basic flow: Choose available power-up → server checks mode/role/balance → charge or reject → apply effect.

UC-06: Customize avatar

Actor: Registered player

Basic flow: Open profile → change options → preview → save → retrieve updated avatar.

8. External Interfaces and Data Requirements

8.1 Client–server boundaries

Interface	Expected data
Authentication API	Username/password or Google credential; user response and signed JWT
Room API	Room settings, room ID, privacy, modes, capacity
Socket.IO gameplay	joinRoom, drawing, guesses, start/end round and score state
WebRTC signaling	Target socket, SDP offer/answer and ICE candidate; microphone audio transmitted by WebRTC
Persistence	User account/avatar/stats; room metadata; reward records
AI word service	Word mode/genre request and candidate word choices with fallback

8.2 Data protection

User credentials must not be exposed to other players. The server, not the client, determines sensitive state such as the secret word and authoritative scores. Socket-level host actions should verify host status. The original design defines a GameSession model for final match records; implementation and persistence are separate verification concerns.

9. Requirements Traceability and Prioritization

The traceability mapping is FR_1–FR_24 ↔ US_1–US_30 in Section 4, and the associated compact acceptance criteria appear in Section 5. FR_5, FR_6, FR_14 and FR_17 each correspond to multiple user stories. The team's later status report considers FR_16 and FR_19 unfinished and the remaining requirements complete; this does not replace testing of each acceptance condition.

Priority group	Examples	Scope rationale
Core access and game loop	FR_1, FR_2, FR_4, FR_5, FR_7–FR_9, FR_23–FR_24	Needed to start, play and complete a match
Engagement and progression	FR_3, FR_10–FR_13, FR_21	Improves retention and player identity
Extended modes and social tools	FR_6, FR_14–FR_20, FR_22	Adds customization, deduction and moderation; can be phased

10. Scope, Assumptions, Constraints and Open Issues

The report records FR_16 (drawer-only reference search) and FR_19 (Extra Clues) as not completed. Other features are classified as done by the team, but an acceptance-test pass/fail record for all 30 stories has not been supplied. Historical audit documents describe an older codebase; they should not override later implementation records without verification.

Additional open points: the original spreadsheet's complete acceptance-criteria text is visible only in screenshots, some cells are truncated; no measured latency or load results are included; finished match-history persistence must be checked separately; voice connectivity can fail on some networks without TURN relay configuration.

11. Requirements Validation and Change Control

11.1 Validation method

The SRS process followed stakeholder questions, role-based user stories, functional requirements and acceptance criteria. The team used the INVEST principles as a writing guide. A requirement should be accepted only after its criteria are tested against a running build; the completion status is not itself proof of acceptance.

11.2 Historical audit versus final scope

The supplied acceptance_criteria_audit and priority audit record findings from an earlier source snapshot, including concerns about reward history, global rankings and voice chat. Later project documents include modules for these features. This SRS keeps both as separate historical evidence and does not treat the older audit as a final test verdict.

11.3 Change record

Change	Basis	Recorded result
FR_16 reference search	Scope decision reported by team	Not Done
FR_19 Extra Clues	Scope decision reported by team	Not Done
Remaining FR entries	Latest team status update	Done (team-reported; verify acceptance criteria)

12. References

[1] Group 7 ID 102_115_160 SRS Presentation(1).pdf — stakeholder roles, process, SRS framing.

[2] SPL-II SRS Presentation_20260715_133220_0000(1).pdf — presentation of requirements engineering process.

[3] SQUIGGLE (Responses)(1).pdf — Google Form survey export, 19 responses, 28–29 June 2026.

[4] Original Google Sheets screenshots, FR_1–FR_24 / US_1–US_30, supplied in conversation — acceptance criteria and statuses.

[5] Group 7 [102, 115, 160].pdf — project proposal and initial non-functional goals.

[6] acceptance_criteria_audit(1).md — historical acceptance-criteria audit.

[7] srs_priority_and_fr_audit(1).md — historical priority/requirement critique.

[8] Group 7 ID 102_115_160 Design Presentation (1)(1).pdf — design references.

[9] 3layerarchi(squiggle)(1).pdf — system architecture reference.